

Practical Nonidentification Example

Mark Fulginiti

2026-06-30

A Simple Example of Practical Nonidentification

The central idea is that two different latent causal mechanisms can produce the same observed treatment–outcome distribution while implying very different causal effects.

Consider a simple setting with:

$$A \in \{0, 1\},$$

where $A = 1$ indicates treatment, and

$$Y \in \{0, 1\},$$

where $Y = 1$ indicates an adverse outcome.

Let Y^0 denote the outcome that would occur without treatment and Y^1 the outcome that would occur under treatment. The average treatment effect on the risk scale is

$$\text{ATE} = E(Y^1 - Y^0).$$

A negative ATE means that treatment reduces the risk of the adverse outcome and is therefore beneficial. A positive ATE means that treatment increases risk and is harmful.

Suppose the observed data satisfy

$$P(A = 1) = 0.5,$$

and

$$P(Y = 1 \mid A = 0) = 0.4, \quad P(Y = 1 \mid A = 1) = 0.6.$$

The observed risk difference is therefore

$$P(Y = 1 \mid A = 1) - P(Y = 1 \mid A = 0) = 0.6 - 0.4 = 0.2.$$

A naïve comparison would suggest that treatment increases the risk of the adverse outcome by 0.2. However, the same observed distribution can arise from two very different causal mechanisms.

Mechanism 1: Treatment is harmful

Suppose treatment is randomized, so there is no confounding:

$$A \perp (Y^0, Y^1).$$

Let

$$P(Y^0 = 1) = 0.4, \quad P(Y^1 = 1) = 0.6.$$

Because treatment is randomized,

$$P(Y = 1 \mid A = 0) = P(Y^0 = 1) = 0.4, \quad P(Y = 1 \mid A = 1) = P(Y^1 = 1) = 0.6.$$

The average treatment effect is

$$\text{ATE}_1 = E(Y^1 - Y^0) = 0.6 - 0.4 = 0.2.$$

Thus, under Mechanism 1, treatment is harmful. The treated group has a higher risk because treatment itself increases the probability of the adverse outcome.

Mechanism 2: Treatment is beneficial, but sicker patients are more likely to receive it

Now suppose there is an unobserved severity variable

$$U \in \{0, 1\},$$

where $U = 0$ denotes lower severity and $U = 1$ denotes higher severity.

Assume

$$U \sim \text{Bernoulli}(0.5),$$

so that

$$P(U = 0) = P(U = 1) = 0.5.$$

Suppose treatment assignment depends strongly on severity:

$$P(A = 1 \mid U = 0) = 0.2, \quad P(A = 1 \mid U = 1) = 0.8.$$

Thus, patients with greater latent severity are much more likely to receive treatment.

The marginal treatment probability is

$$\begin{aligned} P(A = 1) &= P(A = 1 \mid U = 0)P(U = 0) + P(A = 1 \mid U = 1)P(U = 1) \\ &= 0.2(0.5) + 0.8(0.5) = 0.5. \end{aligned}$$

Therefore, this mechanism reproduces the same observed treatment prevalence.

Using Bayes' rule,

$$P(U = 1 | A = 1) = \frac{P(A = 1 | U = 1)P(U = 1)}{P(A = 1)} = \frac{0.8(0.5)}{0.5} = 0.8.$$

Similarly,

$$P(U = 1 | A = 0) = \frac{P(A = 0 | U = 1)P(U = 1)}{P(A = 0)} = \frac{0.2(0.5)}{0.5} = 0.2.$$

Thus, 80% of the treated group is severely ill, whereas only 20% of the untreated group is severely ill.

Now define the potential-outcome probabilities within each severity stratum.

For lower-severity patients,

$$P(Y^0 = 1 | U = 0) = \frac{4}{15}, \quad P(Y^1 = 1 | U = 0) = \frac{1}{15}.$$

For higher-severity patients,

$$P(Y^0 = 1 | U = 1) = \frac{14}{15}, \quad P(Y^1 = 1 | U = 1) = \frac{11}{15}.$$

Within the lower-severity group, the treatment effect is

$$P(Y^1 = 1 | U = 0) - P(Y^0 = 1 | U = 0) = \frac{1}{15} - \frac{4}{15} = -\frac{3}{15} = -0.2.$$

Within the higher-severity group, the treatment effect is

$$P(Y^1 = 1 | U = 1) - P(Y^0 = 1 | U = 1) = \frac{11}{15} - \frac{14}{15} = -\frac{3}{15} = -0.2.$$

Treatment therefore reduces the adverse-outcome probability by 0.2 in both severity groups. Consequently,

$$ATE_2 = E(Y^1 - Y^0) = -0.2.$$

Under Mechanism 2, treatment is beneficial.

The observed treated risk under Mechanism 2:

Among treated patients,

$$\begin{aligned} P(Y = 1 | A = 1) &= \sum_{u=0}^1 P(Y^1 = 1 | U = u)P(U = u | A = 1). \\ &= \frac{1}{15}(0.2) + \frac{11}{15}(0.8) = 0.6. \end{aligned}$$

The observed untreated risk under Mechanism 2:

Among untreated patients,

$$\begin{aligned}
P(Y = 1 \mid A = 0) &= \sum_{u=0}^1 P(Y^0 = 1 \mid U = u)P(U = u \mid A = 0) \\
&= \frac{4}{15}(0.8) + \frac{14}{15}(0.2) = 0.4.
\end{aligned}$$

Thus, Mechanism 2 produces exactly the same observed probabilities:

$$P(A = 1) = 0.5, \quad P(Y = 1 \mid A = 0) = 0.4, \quad P(Y = 1 \mid A = 1) = 0.6.$$

Comparison of the two mechanisms

The two mechanisms produce the same observed treatment–outcome distribution:

$$P_{M_1}(A, Y) = P_{M_2}(A, Y).$$

However, they imply opposite causal effects:

$$\text{ATE}_{M_1} = 0.2, \quad \text{ATE}_{M_2} = -0.2.$$

Under Mechanism 1, treatment is harmful.

Under Mechanism 2, treatment is beneficial, but treated patients appear to have worse outcomes because they are much more severely ill before treatment.

The observed association is the same:

$$P(Y = 1 \mid A = 1) - P(Y = 1 \mid A = 0) = 0.2,$$

but its causal interpretation is completely different.

Why this demonstrates nonidentification:

The causal effect is identified only if it is uniquely determined by the observed-data distribution.

Here, the same observed distribution is compatible with two different values of the causal estimand:

$$P_{M_1}(A, Y) = P_{M_2}(A, Y),$$

while

$$\psi(M_1) \neq \psi(M_2).$$

In particular,

$$\psi(M_1) = 0.2, \quad \psi(M_2) = -0.2.$$

Therefore, the causal effect cannot be recovered uniquely from the observed distribution of (A, Y) alone.

The observed data tell us that treated patients have worse outcomes, but they do not tell us whether treatment caused the worse outcomes or sicker patients were more likely to receive a beneficial treatment.

Without observing or adequately measuring latent severity, these explanations cannot be distinguished from the observed treatment–outcome distribution alone.

Connection to practical nonidentification

In exact mathematical nonidentification, two mechanisms produce precisely the same observed-data distribution:

$$P_{M_1}(O) = P_{M_2}(O),$$

while implying different causal effects:

$$\psi_{M_1} \neq \psi_{M_2}.$$

In a realistic simulation, it may be unnecessary to make the observed distributions exactly equal. Instead, the mechanisms may be calibrated so that

$$P_{M_1}(O) \approx P_{M_2}(O),$$

while

$$\psi_{M_1} \neq \psi_{M_2}.$$

If the observed distributions are sufficiently similar that realistic samples cannot reliably distinguish them, then the problem is practically nonidentified.

The binary example gives the simplest exact construction. My thesis extends the same principle to a more complex observed-data structure involving longitudinal biomarkers, treatment, censoring, and survival:

$$O = (X, \bar{W}, A, \tilde{T}, \Delta).$$

Two latent severity mechanisms may generate similar distributions of the observed data while implying different RMST effects:

$$P_{M_1}(O) \approx P_{M_2}(O),$$

but

$$\psi_{M_1}(\tau) \neq \psi_{M_2}(\tau).$$

The core point is therefore:

Observational similarity does not imply causal similarity. Different latent mechanisms can generate essentially the same observed evidence while implying opposite treatment effects.